

The Scientist and Engineer's Guide to Digital Signal Processing

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Chapter 13: Continuous Signal Processing

The Fourier Series

This brings us to the last member of the Fourier transform family: the *Fourier series*. The time domain signal used in the Fourier series is *periodic* and *continuous*. Figure 13-10 shows several examples of continuous waveforms that repeat themselves from negative to positive infinity. Chapter 11 showed that periodic signals have a frequency spectrum consisting of **harmonics**. For instance, if the time domain repeats at 1000 hertz, the frequency spectrum will contain a first harmonic at 1000 hertz, a second harmonic at 2000 hertz, a third harmonic at 3000 hertz, and so forth. The first harmonic, i.e., the frequency that the time domain repeats itself, is also called the **fundamental frequency**. This means that the frequency spectrum can be viewed in two ways: (1) the frequency spectrum is *continuous*, but zero at all frequencies except the harmonics, or (2) the frequency spectrum is *discrete*, and only *defined* at the harmonic frequencies. In other words, the frequencies between the harmonics can be thought of as having a value of zero, or simply not existing. The important point is that they do not contribute to forming the time domain signal.

The Fourier series **synthesis equation** creates a continuous periodic signal with a fundamental frequency, f , by adding scaled cosine and sine waves with frequencies: f , $2f$, $3f$, $4f$, etc. The amplitudes of the cosine waves are held in the variables: a_1 , a_2 , a_3 , a_4 , etc., while the amplitudes of the sine waves are held in: b_1 , b_2 , b_3 , b_4 , and so on. In other words, the "a" and "b" coefficients are the real and imaginary parts of the frequency spectrum, respectively. In addition, the coefficient a_0 is used to hold the DC value of the time domain waveform. This can be viewed as the amplitude of a cosine wave with zero frequency (a constant value). Sometimes is grouped with the other "a" coefficients, but it is often handled separately because it requires special calculations. There is no b_0 coefficient since a sine wave of zero frequency has a constant value of zero, and would be quite useless. The synthesis equation is written:

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi f t n) - \sum_{n=1}^{\infty} b_n \sin(2\pi f t n)$$

EQUATION 13-4

The Fourier series synthesis equation. Any periodic signal, $x(t)$, can be reconstructed from sine and cosine waves with frequencies that are multiples of the fundamental, f . The a_n and b_n coefficients hold the amplitudes of the cosine and sine waves, respectively.

The corresponding **analysis equations** for the Fourier series are usually written in terms of the *period* of the waveform, denoted by T , rather than the fundamental frequency, f (where $f = 1/T$). Since the time domain signal is periodic, the sine and cosine wave correlation only needs to be evaluated over a single period, i.e., $-T/2$ to $T/2$, 0 to T , $-T$ to 0, etc. Selecting different limits makes the mathematics different, but the final answer is always the same. The Fourier series analysis equations are:

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos\left(\frac{2\pi t n}{T}\right) dt$$

EQUATION 13-5

Fourier series analysis equations. In these equations, $x(t)$ is the time domain signal being decomposed, a_0 is the DC component, a_n & b_n hold the amplitudes of the cosine and sine waves, respectively, and T is the period of the signal, i.e., the reciprocal of the fundamental frequency.

$$b_n = \frac{-2}{T} \int_{-T/2}^{T/2} x(t) \sin\left(\frac{2\pi t n}{T}\right) dt$$

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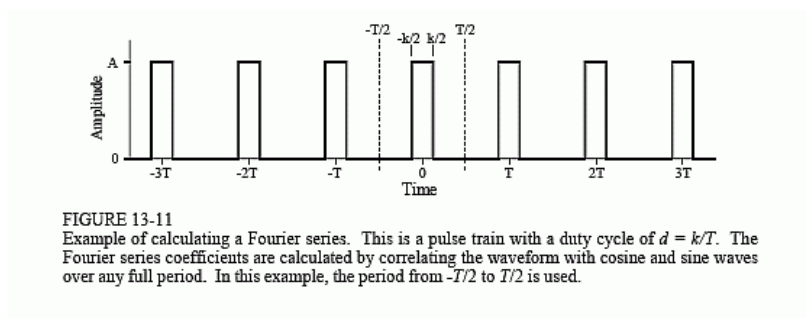
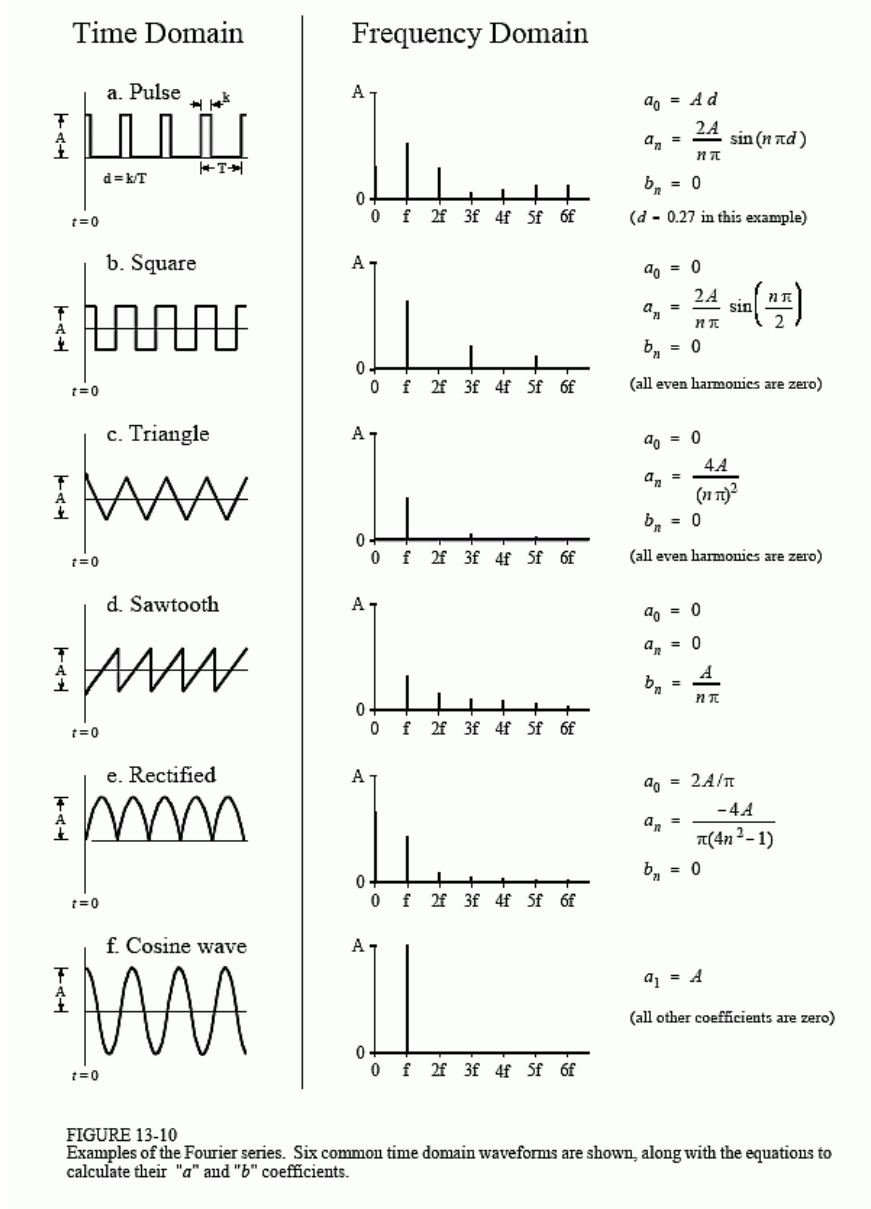


Figure 13-11 shows an example of calculating a Fourier series using these equations. The time domain signal being analyzed is a *pulse train*, a square wave with unequal high and low durations. Over a single period from $-T/2$ to $T/2$, the waveform is given by:

$$x(t) = A \quad \text{for } -k/2 \leq t \leq k/2$$

$$x(t) = 0 \quad \text{otherwise}$$

The *duty cycle* of the waveform (the fraction of time that the pulse is "high") is thus given by $d = k/T$. The Fourier series coefficients can be found by evaluating Eq. 13-5. First, we will find the DC component, a_0 :

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad (\text{start with Eq. 13-5})$$

$$a_0 = \frac{1}{T} \int_{-k/2}^{k/2} A dt \quad (\text{plug in the signal})$$

$$a_0 = \frac{A k}{T} \quad (\text{evaluate the integral})$$

$$a_0 = A d \quad (\text{substitute: } d = k/T)$$

This result should make intuitive sense; the DC component is simply the average value of the signal. A similar analysis provides the "a" coefficients:

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos\left(\frac{2\pi t n}{T}\right) dt \quad (\text{start with Eq. 13-4})$$

$$a_n = \frac{2}{T} \int_{-k/2}^{k/2} A \cos\left(\frac{2\pi t n}{T}\right) dt \quad (\text{plug in the signal})$$

$$a_n = \frac{2A}{T} \left[\frac{T}{2\pi n} \sin\left(\frac{2\pi t n}{T}\right) \right] \bigg|_{-k/2}^{k/2} \quad (\text{evaluate the integral})$$

$$a_n = \frac{2A}{n\pi} \sin(\pi n d) \quad (\text{reduce})$$

The "b" coefficients are calculated in this same way; however, they all turn out to be *zero*. In other words, this waveform can be constructed using only cosine waves, with no sine waves being needed.

The "a" and "b" coefficients will change if the time domain waveform is shifted left or right. For instance, the "b" coefficients in this example will be zero *only* if one of the pulses is centered on $t = 0$. Think about it this way. If the waveform is *even* (i.e., symmetrical around $t = 0$), it will be composed solely of *even* sinusoids, that is, cosine waves. This makes all of the "b" coefficients equal to zero. If the waveform is *odd* (i.e., symmetrical but opposite in sign around $t = 0$), it will be composed of *odd* sinusoids, i.e., sine waves. This results in the "a" coefficients being zero. If the coefficients are converted to polar notation (say, M_n and θ_n coefficients), a shift in the time domain leaves the magnitude unchanged, but adds a linear component to the phase.

To complete this example, imagine a pulse train existing in an electronic circuit, with a frequency of 1 kHz, an amplitude of one volt, and a duty cycle of 0.2. The table in Fig. 13-12 provides the amplitude of each harmonic contained in this waveform. Figure 13-12 also shows the synthesis of the waveform using only the *first fourteen* of these harmonics. Even with this number of harmonics, the reconstruction is not very good. In mathematical jargon, the Fourier series *converges* very *slowly*. This is just another way of saying that sharp edges in the time domain waveform results in very high frequencies in the spectrum. Lastly, be sure and notice the overshoot at the sharp edges, i.e., the Gibbs effect discussed in Chapter 11.

An important application of the Fourier series is electronic **frequency multiplication**. Suppose you want to construct a very stable sine wave oscillator at 150 MHz. This might be needed, for example, in a radio transmitter operating at this frequency. High stability calls for the circuit to be *crystal controlled*. That is, the frequency of the oscillator is determined by a resonating quartz crystal that is a part of the circuit. The problem is, quartz crystals only work to about 10 MHz. The solution is to build a crystal controlled oscillator operating somewhere between 1 and 10 MHz, and then *multiply* the frequency to whatever you need. This is accomplished by *distorting* the sine wave, such as by clipping the peaks with a diode, or running the waveform through a squaring circuit. The harmonics in the distorted waveform are then isolated with band-pass filters. This allows the frequency to be doubled, tripled, or multiplied by even higher integers numbers. The most common technique is to use sequential stages of doublers and triplers to generate the required frequency multiplication, rather than just a single stage. The Fourier series is important to this type of design because it describes the *amplitude* of the multiplied signal, depending on the type of distortion and harmonic selected.

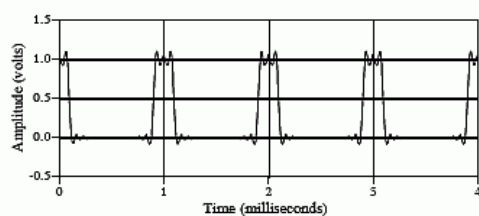


FIGURE 13-12

Example of Fourier series synthesis. The waveform being constructed is a pulse train at 1 kHz, an amplitude of one volt, and a duty cycle of 0.2 (as illustrated in Fig. 13-11). This table shows the amplitude of the harmonics, while the graph shows the reconstructed waveform using only the first fourteen harmonics.

frequency	amplitude (volts)
DC	0.20000
1 kHz	0.37420
2 kHz	0.30273
3 kHz	0.20182
4 kHz	0.09355
5 kHz	0.00000
6 kHz	-0.06237
7 kHz	-0.08649
8 kHz	-0.07568
9 kHz	-0.04158
10 kHz	0.00000
11 kHz	0.03402
12 kHz	0.05046
:	
123 kHz	0.00492
124 kHz	0.00302
125 kHz	0.00000
126 kHz	-0.00297
:	
803 kHz	0.00075
804 kHz	0.00046
805 kHz	0.00000
806 kHz	-0.00046